Foundation Mathematics 1017SCG
Week 2 Workshop Questions

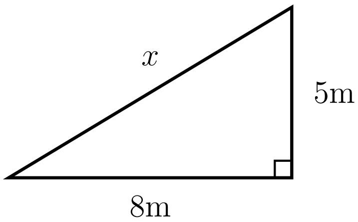
1. Simplify the following. Try to complete this question without using your calculator.
   (a) $\log_4(16)$
   (b) $\log_7(7)$
   (c) $\log_5(125)$
   (d) $\log_9(1)$
   (e) $\log_3\left(\frac{1}{3}\right)$
   (f) $\log_5\left(\frac{1}{25}\right)$
   (g) $\log_2\left(\frac{1}{8}\right)$

2. Simplify the following. Try to complete this question without using your calculator.
   (a) $\log_{12}(6) + \log_{12}(2)$
   (b) $\log_8(2) + \log_8(4)$
   (c) $\log_6(3) + \log_6(12)$
   (d) $\log_4\left(\frac{1}{2}\right) + \log_4(32)$
   (e) $\log_5(8) + \log_5\left(\frac{1}{8}\right)$
   (f) $\log_3(54) - \log_3(6)$
   (g) $\log_5(250) - \log_5(2)$
   (h) $\log_9(90) - \log_9(5) - \log_9(2)$
   (i) $\dfrac{\log_6(x^5)}{\log_6(x)}$
   (j) $\dfrac{\log_4(\sqrt{x})}{\log_4(x)}$

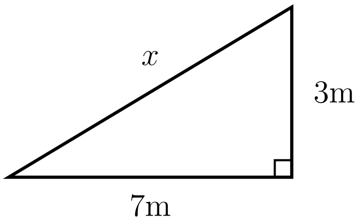
3. Evaluate the following using your calculator, rounding your answer to 4 decimal places.
   (a) $\log_{10}(123)$
   (b) $\ln(40)$
   (c) $\log_5(60)$
   (d) $\log_7(25)$
   (e) $\log_2(18)$
   (f) $\log_9(2)$

4. Find the length of side x in each of the following triangles. Where necessary, round your answer to two decimal places.

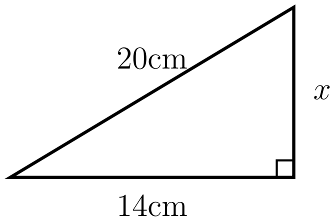
(a)



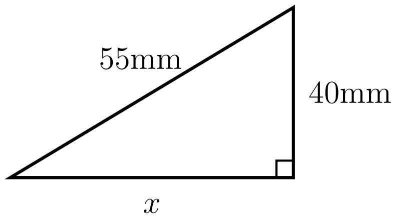
(b)



(c)

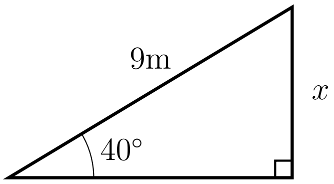


(d)

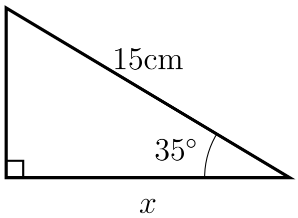


5. Find the length of side x in each of the following triangles. Where necessary, round your answer to two decimal places.

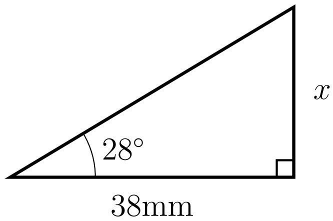
(a)



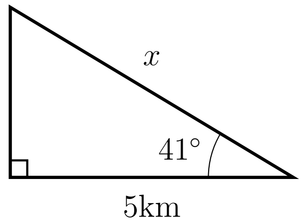
(b)



(c)

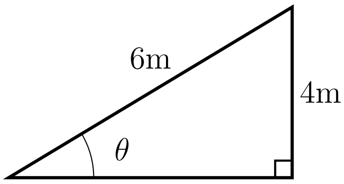


(d)

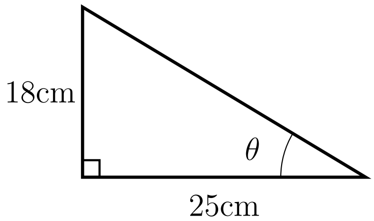


6. Find the size of angle θ in each of the following triangles. Where necessary, round your answer to two decimal places.

(a)

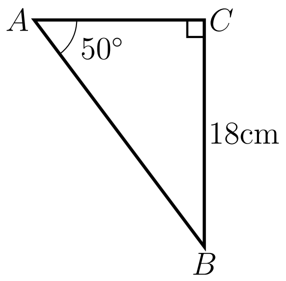


(b)

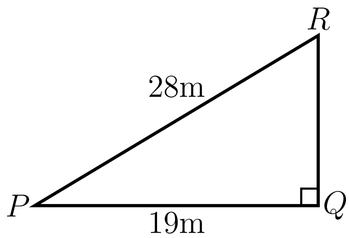


7. Find all unknown side lengths and angles in the following triangles. Where necessary, round your answer to two decimal places.

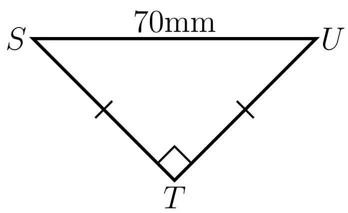
(a)



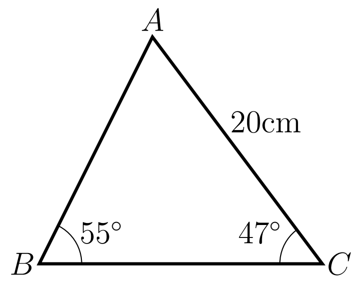
(b)



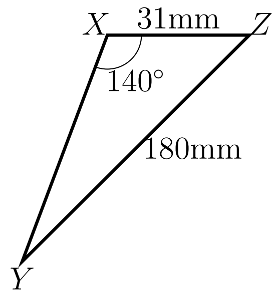
(c)



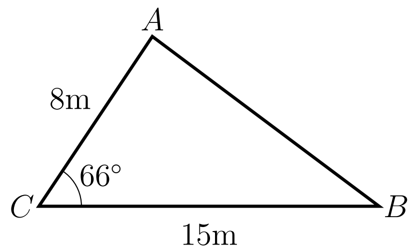
(d)



(e)



(f)



8. Find all unknown side lengths and angles in the following triangles.
   (a) In triangle ABC, $a = 17$mm, $b = 9$mm, angle $C = 118°$
   (b) In triangle ABC, $a = 19$cm, $b = 16$cm, angle $A = 56°$
   (c) In triangle ABC, $a = 23$cm, $b = 12$cm, $c = 28$cm

9. A 4m ladder is leaning against a wall. The vertical distance from the top of the ladder to the floor is twice the horizontal distance from the base of the ladder to the wall. Find the horizontal distance between the base of the ladder and the wall. Assume that the wall is perpendicular to the floor.

10. A hockey goal is 3m wide. When a player is 7.2m from one goal post and 5.8m from the other they take a shot at the goal. Within what angle must the shot be made to score the goal?

11. Two cars depart from the same point at noon. The angle between the two cars' directions is 78°. One car travels at 100km/h and the other travels at 80km/h. What is the distance between the two cars at 2pm?